

Hope Hospital

AI-Assisted SEO Operating System

A complete, modular SEO automation system — phases, workflows, commands, and how it works

hopehospital.com · Built & documented 21 May 2026 · Built with VS Code + Claude

1. What This Is

This is **not** a one-off "morning SEO checklist." It is a modular **operating system** for managing the hospital's entire search presence: technical SEO, GEO (Generative Engine Optimization / AI-search), local SEO, Google Search Console analysis, Google Business Profile, content strategy, internal linking, schema, topical authority, the Ayushman Bharat cluster, reporting, and automation.

The whole system lives under the `seo-system/` folder and is excluded from the public website via `.vercelignore` — so internal SEO operations files are **never served** at hopehospital.com.

2. Core Philosophy

Deterministic scripts find the facts · Claude makes the judgments · a human (and a doctor, for medical claims) approves.

- **Right tool for each job.** Scripts do exact, repetitive checks reliably and free; the AI does prioritisation and content judgment.
- **One workflow at a time.** Build and trust one workflow before scaling — no random tools.
- **YMYL discipline.** It is a hospital. Accuracy and doctor review beat speed and scale. Nothing medical ships without sign-off.
- **Single source of truth.** `master-context.md` holds the hospital profile, priorities, and hard constraints — every workflow reads it.
- **Measure & learn.** Every change is logged and later checked against real Search Console results.

3. How It Works — The Loop

FIND → FIX → MEASURE → LOG → SHIP

The system is built in thin layers, each wrapping the one below:

- **Layer 1 — Engine.** A plain Node script (e.g. `npm run audit`) reads the site files and writes a report. No AI, no internet, runs in ~1 second.
- **Layer 2 — Command.** A Claude slash command (e.g. `/daily-seo`) runs the engine, reads the report + the brain, and returns a prioritised task list.
- **Layer 3 — Schedule.** A morning routine (Claude routine or Windows Task Scheduler) runs it hands-off, so the report is waiting for you.
- **Layer 4 — Data in.** A Search Console export dropped into `seo-system/data/` feeds real ranking data — no API needed.

4. The Six Phases

Phase	What it does	Outcome
1 — Foundation	Deterministic technical-SEO audit engine + the shared "brain"	A trustworthy daily issue list every morning
2 — Reporting & GSC	Reads your Search Console export (no API) for striking-distance & low-CTR opportunities	Data-driven reports; ends hand-writing reports
3 — GEO	Audits AI-search citation readiness and generates structured data	Pages more likely to be cited by ChatGPT / Gemini / Perplexity
4 — Automation	Auto-regenerates the sitemap; one-command morning run; scheduling docs	Sitemap never stale; checks run hands-off
5 — AI Workflows	Internal-link finder, content briefs, Google Business Profile drafts	Topical authority, internal-link coverage, local SEO
6 — Scaling & Governance	KPI trends, outcome measurement (the learning loop), runbook	Knows what actually works; handoff-ready

5. All Commands

Typed in the Claude Code chat box (not the terminal). New commands load when Claude Code starts.

Command	Phase	What it does
<code>/daily-seo</code>	1	Daily SEO plan across the full surface (all ~49 pages), SEO-RICE-ranked
<code>/weekly-report</code>	2	Stakeholder performance report from real GSC data
<code>/geo-audit <page></code>	3	Scores a page's AI-search citation readiness + fixes
<code>/schema-generate <page></code>	3	Generates constraint-safe JSON-LD (refuses to invent doctor data)
<code>/internal-link <page></code>	5	Suggests internal-link anchors + target pages
<code>/content-brief <topic></code>	5	GEO-structured, doctor-reviewable medical content brief
<code>/gbp-post</code>	5	Google Business Profile post + review-reply drafts
<code>/review-outcomes</code>	6	Which past changes actually worked (learning loop)

6. All Engines (Terminal)

Command	What it does
<code>npm run audit</code>	Technical SEO audit (ratings, dates, titles, duplicate titles, alt text, broken links, schema, sitemap)
<code>npm run gsc</code>	Parses the GSC export → striking-distance queries & low-CTR pages
<code>npm run geo -- <page></code>	GEO fact sheet (schema types, FAQ depth, entities, authorship)
<code>npm run links [-- <page>]</code>	Internal-link graph; finds orphan / under-linked pages
<code>npm run sitemap [-- --check]</code>	Regenerates sitemap.xml (<code>--check</code> = preview only)
<code>npm run kpi</code>	KPI trend from the report history
<code>npm run outcome</code>	Flags which logged changes are old enough to evaluate
<code>npm run seo:morning</code>	Runs audit + gsc together

7. The Brain & the Learning Loop

- `project-context/master-context.md` — the single source of truth (hospital profile, priorities, constraints, the SEO-RICE scoring model).
- `project-context/change-ledger.md` — every change logged with a date + hypothesis.
- `project-context/learnings.md` — verdicts on what actually worked, fed back into future plans.
- SEO-RICE** prioritisation: $(\text{search demand} \times \text{position opportunity} \times \text{business value}) \div \text{effort}$ — so task lists are explainable, not guesswork.

8. Hard Constraints (Guardrails)

- **Medical content is doctor-reviewed** before publishing (YMYL / EEAT).
- **No Hindi / Devanagari** text on the website.
- **No new location / branch pages** without an explicit request.
- **No review / aggregateRating schema on blog articles.**
- Google rating is **3.9★ / 402 reviews** — never inflated.
- **No commit or push without explicit approval** (the site is live).

9. Daily & Weekly Routine

Every morning

Run `/daily-seo` → work the top tasks → log each change to `change-ledger.md` → commit (push on approval).

Every week

Export from Search Console → drop the CSVs into `seo-system/data/` → run `/weekly-report`. Then `/review-outcomes` to update `learnings.md`. Run `npm run sitemap -- --check` if pages changed.

10. Real Results Delivered Today

The system did not just get built — it found and fixed real problems on the live site:

Issue (found by the system)	Fix shipped
Website claimed 4.5★ while Google shows 3.9★ (structured-data risk)	Corrected to 3.9★ / 402 across 24 pages
3 cancer pages competing for the same query (cannibalization)	Differentiated; oncology now targets "free cancer treatment"
<code>/services</code> breadcrumb link returned 404 on 9 pages	Added redirect → /departments
Stray "3.8/5" rating text on a blog page	Corrected to 3.9/5
Sitemap hand-maintained & slightly stale	Auto-generated from real pages
Ayushman surgery-list page (top growth driver) had only 1 inbound link	Added links from home + empanelments + sibling blog → 4 inbound

11. Folder Layout

```
seo-system/  
├── project-context/ — the brain: master-context, change-ledger, learnings  
├── scripts/ — deterministic engines (audit, gsc, geo, links, sitemap, kpi, outcome)  
├── reports/ — dated audit / GSC / KPI / outcome reports  
├── automation/ — scheduling docs  
├── gbp/ — Google Business Profile drafts  
├── docs/ — this document  
├── data/ — raw GSC exports (gitignored)  
└── .claude/commands/ — the slash-command workflows
```

12. How to Operate It Going Forward

- **Restart Claude Code** to load any newly added commands.
- **Operate, don't keep building** — run `/daily-seo` daily and work the tasks it surfaces.
- **Feed it data** — a weekly GSC export unlocks real opportunities and outcome measurement.
- **Trust the loop** — in ~2 weeks, `/review-outcomes` will show whether each change actually moved rankings.

